Bing Han, Ph.D.

Los Angeles, CA · Open to relocation (Bay Area / Boston / San Diego) · (+1) 475-655-9070 · hanbingbing0102@gmail.com · binghan-bh.github.io

PROFESSIONAL SUMMARY

Cancer biologist with 10+ years in colorectal cancer and 5+ years leading preclinical programs end to end — from target and mechanism through biomarker strategy to in vivo efficacy. Led a 106-analog small-molecule campaign with medicinal chemistry partners, advancing 10 compounds into efficacy studies; contributed to preclinical validation of a KRAS-targeted mRNA vaccine with a biotech collaborator. **Designs and runs the omics analysis as well as the biology** — scRNA-seq, scATAC-seq, bulk RNA-seq and ChIP-seq in R and Python. First or co-first author on 5 papers; co-author on two *Nature Communications* and two *Science Advances* papers.

CORE TECHNICAL SKILLS

- **Drug discovery and pharmacology:** small-molecule inhibitor campaigns and SAR-guided analog triage; tiered biomarker funnels (target engagement, mechanism of action, cell killing, immune readouts); anti-PD-1 and combination efficacy studies; mRNA vaccine evaluation.
- **In vivo tumor models:** Apc-driven, AOM/DSS-driven and Kras/Apc/Trp53-mutant colorectal models; Kmt2d genetic models; syngeneic and xenograft (CDX/PDX); microbiota colonization and bone loss models. Study design and response assessment.
- **Omics and data analysis:** scRNA-seq and scATAC-seq (Seurat, Signac); bulk RNA-seq, ChIP-seq, m6A-seq, ribosome profiling; TCGA and DepMap; pathway and gene-set analysis. R and Python; reproducible, version-controlled workflows with automated testing.
- **Immune and tissue biomarkers:** flow cytometry, CyTOF immune profiling, IHC/IF, tumor immune microenvironment characterization.
- **Molecular and biochemical methods:** CRISPR gene editing, CRISPR/RNAi screening, qPCR, co-IP, Western blot, immunofluorescence.
- **Cellular systems:** intestinal organoids, cancer cell lines, primary bone marrow cultures, viability and drug sensitivity assays.
- **Disease biology:** colorectal cancer, KRAS/MAPK signaling, inflammation and microbiome-driven tumorigenesis, intestinal stemness, epigenetic and RNA regulation.

RESEARCH EXPERIENCE

Postdoctoral Scholar, Cancer Biology

University of California, Los Angeles | Los Angeles, CA | Apr 2021 – Sep 2026

Small-molecule screening and in vivo validation (project lead)

- Led a screening program for BMI1 inhibitors in colorectal and squamous cell carcinoma, working with medicinal chemistry collaborators on 106 analogs designed from the reference compound PTC596 (unesbulin).
- Built a tiered biomarker funnel so each round of analogs received a go/no-go in days rather than months: target engagement (BMI1, H2AK119ub), mechanism of action (tubulin polymerization; CDK1/2 activation via p38 MAPK), tumor cell killing, and immune chemokine readouts.
- Advanced 10 compounds into in vivo efficacy studies; **2 matched the reference compound's tumor suppression**.

Microbiome-driven tumor immune remodeling in colorectal cancer (project lead)

- Led a program showing that *Fusobacterium nucleatum* drives colorectal cancer through KMT2D-dependent inflammatory reprogramming of the tumor immune microenvironment (*Science Advances*, under revision).
- Established in vivo models combining microbiota exposure, Kmt2d genetic perturbation, immune profiling and therapeutic intervention; showed that targeting inflammation suppressed tumor progression and **enhanced anti-PD-1 efficacy in vivo**.
- Profiled the tumor microenvironment by flow cytometry, CyTOF and IHC/IF, and led the scRNA-seq and scATAC-seq analysis to identify myeloid programs and MDSC/macrophage enrichment; validated an IKK2–NF-κB/p65–KMT2D regulatory axis by ChIP-qPCR at inflammatory gene promoters (FOSL1, TNF).

KRAS-targeted mRNA vaccine (industry collaboration)

- Contributed to preclinical validation of a KRAS mutation–targeted mRNA vaccine in Kras/Apc/Trp53-mutant colorectal cancer mouse models, in collaboration with a biotech partner; supported in vivo study execution and therapeutic response assessment.

Bone metabolism and environmental exposure (project lead)

- Followed up a clinical observation with complementary in vitro and in vivo models to identify environmental factors influencing bone metabolism, and defined the signaling mechanisms involved in bone remodeling (manuscript in preparation).

Visiting Scholar, Cancer Biology

Yale University | New Haven, CT | Sep 2019 – Nov 2020

- Studied epitranscriptomic regulation of pancreatic cancer metastasis and characterized cancer-associated mutations in RNA methylation enzymes (co-author, *Science Advances* 2023).

Ph.D. Researcher, Cancer Biology

Zhejiang University | Hangzhou, China | Sep 2015 – Dec 2020

- Discovered that the m6A reader YTHDF1 is upregulated in colorectal cancer and amplifies Wnt-driven intestinal stemness through translational control (first author, *EMBO Reports* 2020).

INDEPENDENT COMPUTATIONAL PROJECTS

KMT2D dependency across tumour lineages · 2025 – present

- Integrated 19 KMT2D perturbation contrasts from 14 GEO series (99 samples, 6 tissues) with DepMap CRISPR screens across 1,361 cell lines to ask what sets a tumour cell's dependency on a chromatin regulator.
- Found the requirement is set by epithelial differentiation state rather than lineage or mutation, with two independent data types converging on the same axis; reported the ceiling on predictive power explicitly rather than presenting a weak signature.

Selection or adaptation under neoadjuvant PD-1 blockade · 2026

- Decomposed the response of 22 paired pre- and post-treatment colorectal tumours (975,000 cells, 1.3 billion matrix entries, streamed in one pass) into cell-type composition shifts versus expression change within cell states; about 60% of the gene-level change sits within cell states, led by T cells in 12 of 22 patients.
- Reported that the pre-registered specificity control did not pass — adjacent normal tissue splits the same way and changes as much — so the measurement is not attributed to the tumour or to the treatment. Separately quantified what the conventional analysis costs: a cell-level test calls 3,324 genes significant after Bonferroni correction on data built to contain no effect, where a patient-level test on the same cells calls none; reproduced in an independent cohort.
- Both projects are pre-registered before each analysis is run, version-controlled and unit tested, and record the claims my own controls refuted. I designed the studies and the statistical approach and interpreted the results; implementation was AI-assisted. Code available on request.

EDUCATION

Ph.D., Cancer Biology · Zhejiang University, Hangzhou, China · 2015–2020

B.Med., Preventive Medicine · Wuhan University, Wuhan, China · 2010–2015

AWARDS

Poster Presentation Winner · Science Day, Division of Regenerative and Reconstructive Sciences, UCLA · 2025 and 2026

SELECTED PUBLICATIONS

First or co-first author

- **Han B**, Liu W, Chen Y, et al. *Fusobacterium nucleatum* promotes colorectal cancer via KMT2D-dependent transcriptional reprogramming of tumor immune environment. *Science Advances*. Under revision.
- **Han B**, Yan S, Wei S, et al. YTHDF1-mediated translation amplifies Wnt-driven intestinal stemness. *EMBO Reports*. 2020;21:e49229.
- **Han B**, Wei S, Li F, et al. Decoding m6A mRNA methylation by reader proteins in cancer. *Cancer Letters*. 2021;518:256–265.
- Zhang L*, **Han B***, Xiang J, et al. Silica nanoparticle releases SIRT6-induced epigenetic silencing of follistatin. *Int J Biochem Cell Biol*. 2018;95:27–34. (*co-first)
- **Han B**, Liu W, Chen Y, et al. Modeling *Fusobacterium nucleatum*-driven colorectal cancer in Apc-mutant mice. In: *Fusobacterium nucleatum: Methods and Protocols*.

Selected co-authorships

- Liu W, Cao H, Wang J, Elmusrati A, **Han B**, et al. Histone-methyltransferase KMT2D deficiency impairs the Fanconi anemia/BRCA pathway upon glycolytic inhibition in squamous cell carcinoma. *Nature Communications*. 2024;15:6755.
- Hou J, Gao Y, **Han B**, Yan S, Wei S, Gao X. Nuclear accumulation of YTHDF1 regulates mRNA splicing in the DNA damage response. *Science Advances*. 2025;11:eado7660.
- Yan S, Zhou X, Wu C, **Han B**, et al. Adipocyte YTH N6-methyladenosine RNA-binding protein 1 protects against obesity by promoting white adipose tissue beiging in male mice. *Nature Communications*. 2023;14:1379.
- Perez-Pepe M, Desotell AW, Li H, Li W, **Han B**, et al. 7SK methylation by METTL3 promotes transcriptional activity. *Science Advances*. 2023;9:eade7500.
- Wang F, Jiang M, Zhu L, et al. (**Han B**, co-corresponding author). Lactic acid and lactylation in the progression of pulmonary fibrosis. *BIO Integration*. 2026;7(1):975.

Full publication list: **Google Scholar**